

Token-Level Missingness Training Can Erode Clean-Input Performance in Multimodal Learning

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Abstract

Training for missing modalities can change a multimodal system’s performance even when every input is available. Temporal erasure and whole-modality dropout expose the model to different observations and reconstruction signals, although both are specified by a missing rate. We compare these training protocols through their clean-input accuracy under benign and text-scarce training regimes, using matched nominal rate schedules and a shared fusion architecture with availability-gated reconstruction. Across CMU-MOSI, CMU-MOSEI and CH-SIMS, text-scarce token-level training reduces accuracy by up to 27.7 percentage points relative to benign training; the largest corresponding decrease under modality-level training is 2.2 points. In a rate sweep, the token-level decline is first more than two percentage points at 50–60% text erasure, and subsampling CMU-MOSEI produces larger changes at smaller sample sizes. The result replicates with BERT and RoBERTa; an [UNK]-substitution control produces a 29.6-point decrease. In intervention experiments, modality-level training keeps clean-input accuracy close to the benign-regime reference, while freezing lower layers or reducing the text-encoder learning rate does not recover the token-level result. A separate 40-run control finds a 23.0-point token-level decrease even with auxiliary reconstruction loss removed, compared with 1.1 points under modality-level training. These findings show that matched nominal missing rates can produce different clean-input outcomes within a shared architecture, motivating explicit evaluation of complete inputs alongside missing-input robustness.

Keywords: incomplete multimodal learning, missing modalities, multimodal sentiment analysis, fine-tuning dynamics, clean-input performance

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1. Introduction

Multimodal sentiment systems must remain useful when text, audio or video is unavailable. For example, a transcription service can fail during an otherwise intact audiovisual interaction and later resume. The system must handle both the interruption and the return of complete inputs. Training pipelines prepare for this condition by simulating missingness. One common protocol erases temporal positions within a modality, leaving a partially observed stream; we call it token-level missingness (TOK). Another removes a complete stream; we call it modality-level missingness (MOD). Both are parameterised by a missing rate and are often reported as alternative implementations of the same experimental condition.

Complete-input performance is already a concern in robust multimodal learning [1, 2]. The question here is how it changes across training protocols as simulated missingness increases. A complete-input score from a separately trained model, as in the complete and incomplete settings of EMT-DLFR [3], does not answer that question. We therefore compare models trained under benign and text-scarce missingness on the same unmasked test inputs.

Our central measurement is the *clean-input change*: the difference in complete-input accuracy between text-scarce and benign missingness training. At matched nominal rate schedules, the two protocols separate sharply. token-level training changes clean-input binary accuracy by as much as -27.7 points on CMU-MOSI, while modality-level training decreases it by no more than 2.2 points in the corresponding comparisons. The rate sweep identifies the first token-level declines of more than two percentage points at text-erasure rates of 50–60%, placing the effect inside the range used by published missing-modality studies [3, 4].

The comparison covers data scale, text encoder and token-corruption operator within a shared reconstruction-based fusion architecture. It appears on CMU-MOSI, CMU-MOSEI and CH-SIMS, replicates with BERT and RoBERTa [5, 6], and becomes larger on smaller training sets. Replacing removal from the attention mask with the [UNK] substitution used by EMT-DLFR [3] increases the clean-input change from -20.8 to -29.6 points. The effect therefore persists across two text encoders and two token-corruption operators. A separate five-seed factorial control varies the reconstruction-loss weight under both protocols and training regimes. The protocol contrast persists at zero loss weight, showing that auxiliary reconstruction supervision is not necessary for the contrast in this configuration (Section Appendix

A).

The intervention study identifies training choices associated with retained clean-input accuracy in this architecture. modality-level dropout keeps accuracy close to its benign-training reference on CMU-MOSI and CH-SIMS, whereas freezing the lower text-encoder layers or reducing their learning rate does not recover the accuracy lost under high-rate token-level erasure. The practical implication is to evaluate complete-input accuracy when choosing the training protocol, together with performance under the intended test-time missingness conditions.

This paper makes the following contributions.

- We compare the clean-input changes of two missingness-training protocols and show that they produce substantially different complete-input outcomes (Section 5).
- We characterise the dose-response relationship and locate the first token-level declines of more than two points at 50–60% erasure on the tested rate grids (Section 5).
- We establish the result across three datasets, BERT and RoBERTa text pathways, and both removal and [UNK] token operators (Sections 5 and 6).
- We show that modality-level dropout retains accuracy near its benign reference in the intervention study, while two plasticity-reducing interventions do not recover it (Section 7).

2. Related work

Methods for incomplete multimodal sentiment analysis differ mainly in how they compensate for an absent stream. Reconstruction-based methods predict the missing representation from the survivors, either at the level of low-level features [7, 3] or through generative transport between modality distributions [8]; graph-completion methods propagate available context across utterances [9]; imagination networks learn a direct mapping from present to absent modalities [10]; prompt-based methods condition a frozen backbone on the observed combination [11]; and mixture-of-experts methods route on the availability pattern [12, 13]. These approaches optimize performance under missing inputs. Our study instead measures what their training protocols retain when complete input returns.

Two simulation protocols are in circulation. EMT-DLFR masks a random subset of temporal positions within each stream, arguing that the “entire loss of one or more modalities . . . is less likely to happen in real-world scenarios” [3]; TFR-Net [7] introduced the same construction, and recent work continues to define intra-modal missingness as temporal erasure at a specified rate [4]. Others drop whole modalities independently with a fixed probability [14], and benchmark efforts standardise rate schedules across modalities without treating granularity as a factor [15]. BALM studies optimization imbalance under unequal modality-level missing rates [16], while ANGA protects complete-sample representations when reconstructed samples dominate training [17]. These studies establish that missingness training can reshape optimization. Our comparison evaluates a different experimental axis: temporal erasure versus whole-modality absence at matched nominal rate schedules.

Preserving complete-input performance is a shared goal of robust multimodal learning. Hazarika et al. [1] study modality sensitivity and robust training strategies. MissModal [2] aligns representations of incomplete and complete observations while preserving sentiment inference with complete modalities. M3S [18] studies missing-modality sampling within a meta-learning framework. These studies motivate treating the training distribution and complete-input outcome together. Our contribution is a comparison of temporal-erasure and whole-modality training protocols across nominal rate, sample size, text encoder and corruption operator.

The intervention study compares changing the training protocol with limiting adaptation in the text pathway. Freezing lower layers and lowering the pretrained encoder’s learning rate provide two concrete controls for the latter strategy (Section 7).

3. Setting

3.1. Missingness operators and observation structure

We compare two input-corruption protocols with the same nominal rate vector. An utterance is a triple $x = (x_t, x_a, x_v)$ of variable-length sequences over text, audio and video, paired with a validity mask $v_k \in \{0, 1\}^{T_k}$ marking real (non-padding) positions of modality k . Both protocols are parameterised by the same per-modality rate vector $r \in [0, 1]^3$. token-level missingness (TOK) erases each real position of modality k independently with probability r_k , so the modality remains present but partially corrupted. modality-level missingness (MOD) erases *every* position of modality k with

probability r_k , so the modality is either intact or entirely gone. Both operators retain at least one modality per example: if an initial draw removes every valid position, the modality with the smallest nominal rate is restored in full, with ties resolved in text–audio–video order. Erased content is zeroed and removed from the validity mask. The restoration rule changes the realized rate differently under the two protocols; matching r therefore matches the nominal schedule. For whole-modality dropout and a fixed r , the restoration event has probability $r_t r_a r_v$. With L_k valid positions per modality, its token-level counterpart has probability $\prod_k r_k^{L_k}$. For text, the sampling population is every position marked valid by the tokenizer, including boundary tokens; the removal and [UNK] controls use the same population.

Matching the nominal rate fixes the expected amount of retained input before restoration, but leaves its distribution different. To make this distinction explicit, let Q_k be the fraction of valid positions retained in a stream of length $L_k > 0$. Conditional on a fixed rate r_k , the initial draws satisfy

$$\mathbb{E}[Q_k \mid \text{TOK}] = \mathbb{E}[Q_k \mid \text{MOD}] = 1 - r_k, \quad (1)$$

$$\text{Var}(Q_k \mid \text{TOK}) = \frac{r_k(1 - r_k)}{L_k}, \quad \text{Var}(Q_k \mid \text{MOD}) = r_k(1 - r_k). \quad (2)$$

Equations (1) and (2) describe the draws before restoration. Token-level erasure concentrates longer sequences around a partially observed state, whereas modality-level dropout places all mass on intact and absent states. For example, with $L_k = 10$ and $r_k = 0.7$, both initially retain three positions on average. The probability of preserving the entire stream is nevertheless 0.3^{10} under token-level erasure and 0.3 under modality-level dropout. Thus an equal nominal rate gives very different exposure to intact sequences. This observation motivates examining complete-input performance separately from the amount of input removed during training.

3.2. Training regimes and clean-input measurement

A *regime* is a distribution over rate vectors, and it is the unit at which prior work reports robustness. We use four: U-LO and U-HI draw every r_k uniformly from $[0.1, 0.3]$ and $[0.5, 0.7]$ respectively; T-FRAG draws $r_t \sim \mathcal{U}(0.6, 0.8)$ with the nonverbal streams at $[0.1, 0.3]$; and NV-FRAG uses $[0.0, 0.2]$ for text and $[0.6, 0.8]$ for each nonverbal stream. The first two vary *how much* is missing while holding the modalities symmetric; the last two vary *which* modality is fragile. For the dose-response analysis we additionally use a one-parameter family $\text{T@}\rho$, fixing $r_t = \rho$ with the nonverbal rates at 0.1.

Table 1: Measured observation and reconstruction exposure on the 1,284 CMU-MOSI training utterances. Each condition uses ten masking repetitions with matched nominal rate draws across protocols. Values include the restoration rule. Rec. pairs is the mean number of fully absent sample–modality pairs per utterance, which determines eligibility for reconstruction supervision.

Regime	Protocol	Intact text (%)	Absent text (%)	Rec. pairs
U-lo	token-level	10.66	0.00	0.001
U-lo	modality-level	81.12	18.88	0.585
T-frag	token-level	0.04	3.25	0.033
T-frag	modality-level	30.33	69.67	1.070

The implemented operators produce different observation exposure on real training-sequence lengths (Table 1). We apply each operator to the CMU-MOSI training validity masks for ten repetitions, using rate seeds 31000–31009 and masking seeds 41000–41009. Under text-scarce training, the measured intact-text frequency is 0.04% for token-level erasure and 30.33% for modality-level dropout. Fully absent text occurs in 3.25% and 69.67% of draws, respectively. Across all three streams, the mean number of reconstruction-eligible pairs per utterance is 0.033 under token-level erasure and 1.070 under modality-level dropout. These measurements describe the input and supervision distributions generated by the training implementation, with the model and task outcomes held outside the measurement. They quantify the exposure differences included in the complete-protocol comparison.

Our primary measurement is taken on *unmasked* input and quantifies the complete-input capability retained after missingness training. Let $A(\pi, g)$ be accuracy on complete input for a model trained under regime π with protocol g . We report $\Delta_{\text{clean}}(g) = 100[A(\text{T-FRAG}, g) - A(\text{U-LO}, g)]$: the clean-input accuracy change in percentage points, with A expressed as a fraction. The reference is benign missingness training. Because the test input is identical and complete in both terms, Δ_{clean} excludes test-time missingness difficulty. It is an end-to-end system measurement and does not assign the change to a particular layer.

Each protocol supplies its own benign reference. Consequently, a small Δ_{clean} means that increasing text scarcity has little effect relative to that protocol’s starting point; its absolute accuracy still matters. We therefore report benign accuracy alongside changes and examine regression error as a second outcome. The rate-sweep reference serves a separate purpose: it holds nonverbal corruption fixed while varying only the nominal text rate.

Its changes describe movement along that sweep rather than the contrast between the two randomized training regimes.

Figure 1 summarizes the paired comparison. Training corruption changes between the two fitted models, while the complete test observations and accuracy definition remain fixed.

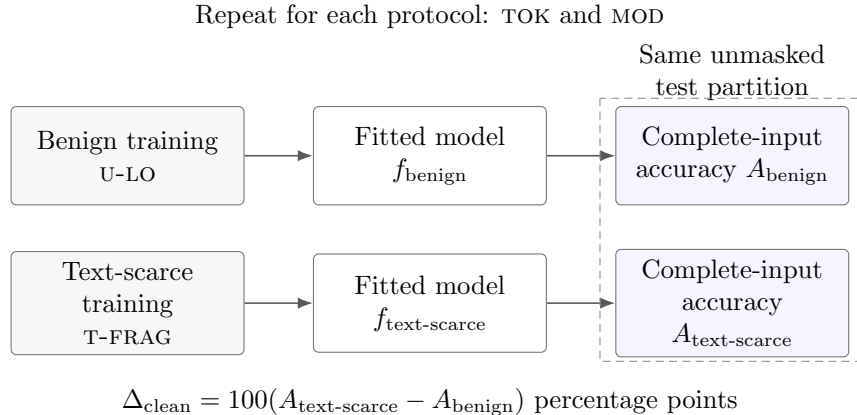


Figure 1: **Clean-input change compares training histories on the same complete test inputs.** The design is repeated separately for token-level erasure (TOK) and modality-level dropout (MOD). Each protocol has its own benign-trained reference; architecture and optimization settings are shared, while model weights are fitted separately. Changes are paired by training-seed identifier before aggregation.

4. Experimental setup

4.1. Datasets and shared architecture

We evaluate on the three standard benchmarks for this task and verify the provenance of every feature file. CMU-MOSI provides 1,284 training, 229 validation and 686 test utterances; CMU-MOSEI provides 16,326, 1,871 and 4,659; and CH-SIMS provides 1,368, 456 and 457. All three are used with the pre-extracted unaligned features distributed by the MMSA toolkit, and each file’s SHA-256 digest is checked against the value published by that project before use. CMU-MOSI and CMU-MOSEI carry sentiment labels in $[-3, 3]$; CH-SIMS uses $[-1, 1]$ and is in Chinese, which additionally lets us test whether our findings depend on the language of the text stream.

We use a shared reconstruction-based fusion architecture to compare the two training protocols. Each modality is encoded by a two-layer transformer with learned positional embeddings and masked mean pooling; per-modality

reconstruction heads predict the pooled representation of an absent modality from the surviving ones, supervised by the pooled representations the same encoders produce on complete input; and a fusion multilayer perceptron consumes the three pooled vectors together with explicit availability flags. The text pathway is the one component we vary: instead of consuming the stored features, a pretrained encoder (**bert-base** for CMU-MOSI and CMU-MOSEI, **bert-base-chinese** for CH-SIMS, **roberta-base** in Section 6) produces per-token representations that enter the *same* modality encoder the stored features would have. Fine-tuning therefore changes how those 768-dimensional vectors are produced within the shared downstream architecture.

All modality encoders project their inputs to dimension $d = 64$ and use four attention heads with feed-forward width $4d$. Masked mean pooling is followed by layer normalization. Residual and feed-forward dropout are 0.1; self-attention dropout in these modality encoders is zero. The pretrained text model retains its own dropout configuration. Each reconstruction head has two hidden layers of width 128, GELU activations and dropout 0.1. The fusion network maps its 195 inputs through widths 128 and 64 to a scalar sentiment prediction. Holding this stack fixed makes the encoder and missingness comparisons interpretable within one shared model family.

4.2. Availability-gated reconstruction objective

Let $\tilde{x}_{ik}$ denote the corrupted stream for example i , a_{ik} indicate whether any valid position survives, and $z_{ik} = E_k(\tilde{x}_{ik}) \in \mathbb{R}^d$ be its pooled encoding. The reconstruction head R_k receives the other streams’ availability-gated encodings and their flags:

$$\hat{z}_{ik} = R_k([(a_{ij}z_{ij})_{j \neq k}, (a_{ij})_{j \neq k}]), \quad (3)$$

$$u_{ik} = a_{ik}z_{ik} + (1 - a_{ik})\hat{z}_{ik}, \quad (4)$$

$$\hat{y}_i = F([u_{it}, u_{ia}, u_{iv}, a_{it}, a_{ia}, a_{iv}]). \quad (5)$$

In Equation (3), brackets denote concatenation. The gate in Equation (4) selects the representation used by the fusion network and scalar head F in Equation (5). A partially observed stream contributes its observed encoding; a fully absent stream contributes its reconstruction. At clean test time all availability flags equal one, so prediction uses the observed encodings directly.

For a batch of B examples, define $m_{ik} = 1 - a_{ik}$ and the detached

complete-input target $z_{ik}^* = \text{sg}(E_k(x_{ik}))$. The training objective is

$$\mathcal{L}_{\text{task}} = \frac{1}{B} \sum_{i=1}^B |\hat{y}_i - y_i|, \quad (6)$$

$$\mathcal{L}_{\text{rec}} = \frac{\sum_{i,k} m_{ik} \|\hat{z}_{ik} - z_{ik}^*\|_2^2 / d}{\max(1, \sum_{i,k} m_{ik})}, \quad (7)$$

$$\mathcal{L} = \mathcal{L}_{\text{task}} + 0.5\mathcal{L}_{\text{rec}}. \quad (8)$$

Equation (8) combines sentiment regression (Equation (6)) with representation reconstruction (Equation (7)). The reconstruction term is zero when the batch contains no fully absent streams. Targets are recomputed with the current encoder parameters on the complete training observations, without back-propagation through the target branch. This branch remains in training mode, including dropout; it is not a separate frozen teacher. Complete observations provide training supervision, while inference uses only the supplied inputs and availability masks.

The comparison is between complete training protocols rather than isolated gradient paths. The reconstruction loss is availability-gated: whole-modality absence activates it whenever a stream is dropped, whereas partial temporal erasure activates it only if no valid position survives. This fixed behaviour is part of the operational difference induced by the two protocols. Before restoration, a stream of length L_k is fully absent with probability $r_k^{L_k}$ under token-level erasure and r_k under modality-level dropout. For longer streams, these probabilities can differ substantially. The reconstruction objective therefore receives different supervision frequencies even though its coefficient and implementation remain fixed.

4.3. Optimization, checkpoint selection and evaluation

Optimisation uses AdamW with two parameter groups. The pretrained encoder is trained at 2×10^{-5} while the randomly initialised fusion stack uses 10^{-3} . Training runs for at most 40 epochs with early stopping on validation mean absolute error, and validation is always performed under the *training* regime. Checkpoints are selected using validation MAE under that regime; test scores are not used for selection. Each selected checkpoint is evaluated on the test conditions described below. Batch size is 32, weight decay is 10^{-4} , and gradients are clipped to norm 1. Early stopping allows

Table 2: Clean-input performance under benign missingness training with BERT. Acc-2 excludes neutral labels; MAE uses all examples. Values are mean $\pm$ standard deviation across five seeds for each dataset and protocol.

Dataset	Training protocol	Acc-2 $\uparrow$	MAE $\downarrow$
CMU-MOSI	token-level	0.835 ± 0.007	0.813 ± 0.024
CMU-MOSI	modality-level	0.845 ± 0.007	0.728 ± 0.019
CMU-MOSEI	token-level	0.842 ± 0.012	0.531 ± 0.004
CMU-MOSEI	modality-level	0.844 ± 0.006	0.525 ± 0.005
CH-SIMS	token-level	0.810 ± 0.011	0.403 ± 0.010
CH-SIMS	modality-level	0.806 ± 0.018	0.411 ± 0.021

eight epochs without an improvement of more than 10^{-5} in validation MAE. Validation averages two corruption draws for each checkpoint evaluation.

Both protocols use a shared masking implementation with random draws on CPU. The reported results evaluate each trained model once on complete test inputs; this evaluation applies no corruption. Archived incomplete-input evaluations average five mask draws. Their hash seeds depended on the Python process, so identical corrupted inputs across separate runs are not guaranteed. The released driver uses stable digest seeds for future runs. Validation also uses missingness, making its sampled masks part of the original checkpoint selection procedure.

Acc-2 is the fraction of correct positive-versus-negative predictions after excluding test examples whose ground-truth sentiment equals zero; predictions above zero are positive. MAE uses all test examples and clips predictions to the dataset’s label range. We report mean $\pm$ standard deviation over three to five training seeds, using population standard deviations to describe the observed runs. Changes are calculated within matched seed identifiers before aggregation. These intervals describe variation across runs, rather than confidence intervals or hypothesis tests. Specifically, for matched seeds $s = 1, \dots, S$, we calculate $\delta_s = 100(A_{s,t\text{-frag}} - A_{s,u\text{-lo}})$, then report $\bar{\delta} = S^{-1} \sum_s \delta_s$ and $\sigma_\delta = [S^{-1} \sum_s (\delta_s - \bar{\delta})^2]^{1/2}$. Pairing preserves the correspondence between run identifiers; it does not imply that independently executed runs used identical validation masks. The sample-size study additionally holds each selected subset fixed across training seeds, so its error bars describe training variation conditional on that subset.

Table 2 reports the benign-regime reference separately for each protocol. This supplies the absolute accuracy and regression error from which the text-scarce comparisons are calculated.

Table 3: Clean-input MAE under benign and text-scarce training with BERT on the full datasets. A positive change indicates higher error. Changes are mean $\pm$ population standard deviation across five matched seeds; the two regime columns show mean MAE. All values use the original sentiment scale.

Dataset	Protocol	Benign	Text-scarce	MAE change $\downarrow$
CMU-MOSI	token-level	0.813	1.323	$+0.510 \pm 0.108$
CMU-MOSI	modality-level	0.728	0.791	$+0.063 \pm 0.040$
CMU-MOSEI	token-level	0.531	0.609	$+0.078 \pm 0.008$
CMU-MOSEI	modality-level	0.525	0.541	$+0.016 \pm 0.011$
CH-SIMS	token-level	0.403	0.443	$+0.039 \pm 0.020$
CH-SIMS	modality-level	0.411	0.391	-0.021 ± 0.026

5. Token-level training erodes clean-input performance

5.1. Clean-input classification and regression

The two protocols produce markedly different clean-input outcomes. Table 4 reports Δ_{clean} , the difference in clean-input accuracy between text-scarce and benign training. The nominal rate schedules are matched across protocols. Under token-level erasure the loss reaches 20.8 points on CMU-MOSI with BERT and 27.7 with RoBERTa. Under modality-level absence the same comparison costs 0.8 and 2.0 points respectively. The direction is consistent across every dataset and both text encoders: token-level erasure changes clean-input accuracy substantially more than modality-level absence.

Because Δ_{clean} is measured on complete input, it is distinct from robustness under a particular test-time mask. A system with $\Delta_{\text{clean}} = -20.8$ has lost complete-input task accuracy after missingness training. Standard evaluations that test a missingness-trained model only on corrupted inputs do not expose this quantity.

Regression error corroborates the classification result in the full-data BERT experiments (Table 3). Text-scarce token-level training increases MAE by 0.510 on CMU-MOSI, 0.078 on CMU-MOSEI and 0.039 on CH-SIMS. With modality-level training, the respective changes are 0.063, 0.016 and -0.021 . Thus the token-level result also appears in distance from the annotated sentiment score, beyond whether a prediction crosses the polarity boundary. On CH-SIMS, modality-level MAE improves even though the accuracy change is small, illustrating that polarity and sentiment magnitude capture different aspects of the output. MAE magnitudes should be compared within a dataset because the label ranges differ.

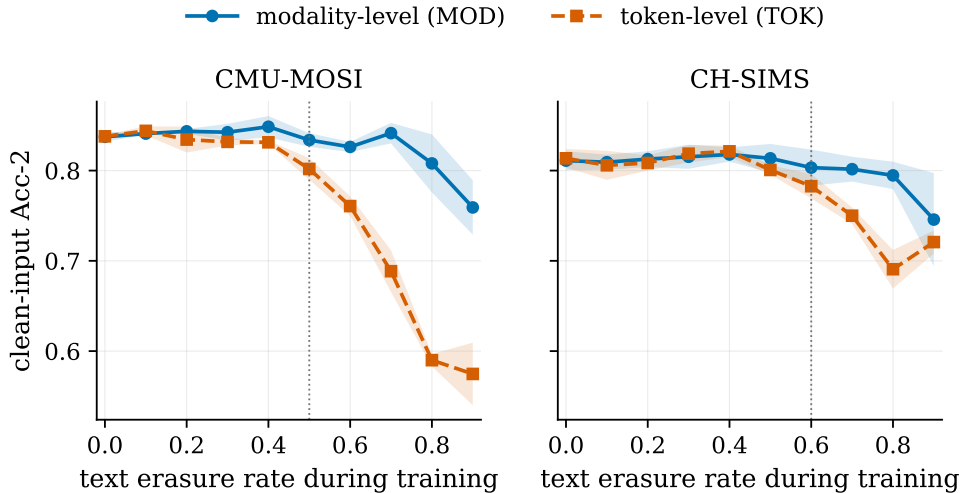


Figure 2: **Training protocols separate as nominal text erasure increases.** Clean-input accuracy after training under token-level erasure (TOK, orange) or modality-level absence (MOD, blue) at a given text erasure rate. Dotted lines mark the first tested token-level rate with more than a two-point decrease from the text-rate-zero reference: 0.5 and 0.6, respectively. Shaded bands show ± 1 standard deviation over three seeds.

5.2. Dependence on the nominal text-erasure rate

The rate sweep locates the range where clean-input performance changes. Figure 2 varies the training text-erasure rate from 0 to 0.9 on CMU-MOSI and CH-SIMS. Below approximately 0.5, the two curves are close. At higher rates they separate sharply: token-level erasure reaches 0.575 accuracy on CMU-MOSI, a drop of 26.3 points from the zero-erasure baseline, while modality-level absence loses 7.8. CH-SIMS has its largest token-level decrease at rate 0.8 (12.3 points), compared with a largest modality-level decrease of 6.5 points at rate 0.9. For descriptive comparison, we define onset as the first tested rate whose mean accuracy is more than two percentage points below that protocol’s rate-zero reference. This occurs at 0.5 on CMU-MOSI and 0.6 on CH-SIMS under token-level training. The criterion identifies points on the tested grid, rather than a fitted change point. Audio and video rates remain 0.1 throughout, including at the text-rate-zero reference.

The onset lies within rates used in prior evaluations. EMT-DLFR evaluates rates from 0.1 to 1.0 on CMU-MOSI and CMU-MOSEI, and up to 0.5 on CH-SIMS [3]; MRCF evaluates intra-modal missingness from 0.0 to 0.9 [4]. The observed onset rates therefore fall within evaluation ranges used

Table 4: Clean-input change Δ_{clean} after text-scarce rather than benign missingness training. The benign Acc-2 column is the token-level reference; modality-level changes use their own benign reference. Values are mean $\pm$ standard deviation of paired-seed changes, with five seeds for full-data BERT and three for subsampled BERT and RoBERTa.

Text encoder	Dataset	benign	clean-input Acc-2 change (pts)	
		Acc-2	token-level	modality-level
BERT	CMU-MOSI	0.835 ± 0.007	-20.8 ± 7.9	-0.8 ± 1.6
	CMU-MOSEI ($n=1284$)	0.839 ± 0.005	-12.7 ± 4.2	-1.7 ± 0.4
	CH-SIMS	0.810 ± 0.011	-4.6 ± 2.3	$+0.1 \pm 2.0$
	CMU-MOSEI	0.842 ± 0.012	-3.2 ± 1.1	-0.7 ± 0.8
RoBERTa	CMU-MOSI	0.853 ± 0.011	-27.7 ± 3.7	-2.0 ± 2.4
	CMU-MOSEI ($n=1284$)	0.838 ± 0.002	-18.1 ± 0.7	-2.2 ± 2.2

in prior studies.

5.3. Unknown-token substitution control

The operator check repeats the comparison with the text substitution used in prior work. Our token-level operator removes a masked position from the attention mask, so the token becomes invisible. EMT-DLFR instead “replaces the original token with the [UNK] token in BERT vocabulary” [3], leaving the position visible but uninformative. Repeating the CMU-MOSI experiment with that substitution in place of removal gives a clean-input change of -29.6 points ($0.828 \rightarrow 0.532$ clean-input accuracy, three seeds) against -20.8 for removal. Substituting an unknown-token embedding at most positions produces a larger clean-input change than making those positions invisible. The observation is therefore not specific to removal from the attention mask.

6. What modulates the clean-input change

6.1. Training-set size

Training-set size strongly modulates the magnitude of the clean-input change. The effect is much larger on CMU-MOSI (20.8 points) than on full CMU-MOSEI (3.2), and those datasets differ in many ways. We therefore examined training-set size through nested CMU-MOSEI subsets while holding features, architecture, protocol and full-training-set normalization statistics fixed. The subsets are nested prefixes of one seeded permutation, shared across training seeds. Figure 3a shows the result: the overall change grows as data shrink, from 3.2 points at 16,326 training examples

Table 5: CMU-MOSI tokenisation audit at a 50-token limit, including special tokens. Length is the mean retained token count; truncated is the percentage whose untruncated length exceeds 50. Lowercased RoBERTa identifiers and masks match the experiment cache on all three partitions.

Tokenizer / input	Training		Test	
	Length	Trunc. (%)	Length	Trunc. (%)
BERT-uncased	14.78	0.55	16.50	2.04
RoBERTa / original	20.49	3.58	22.50	5.54
RoBERTa / lowercase	14.81	0.62	16.54	1.90

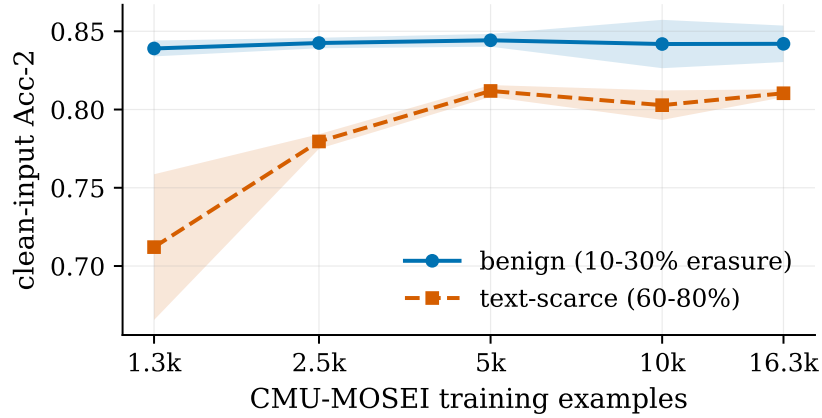
to 12.7 at 1,284, a fourfold increase over a $12.7\times$ reduction in data. Two features of that curve matter. The benign-regime line is flat across the whole range (0.839 to 0.844), so the observed size sensitivity is concentrated in the text-scarce regime. The intermediate values are not strictly monotonic. At CMU-MOSI’s exact training-set size, CMU-MOSEI reaches 12.7 points against CMU-MOSI’s 20.8 points. The within-CMU-MOSEI comparison supports a sample-size association, while the remaining cross-dataset difference also reflects corpus variation.

6.2. Text encoder and corpus variation

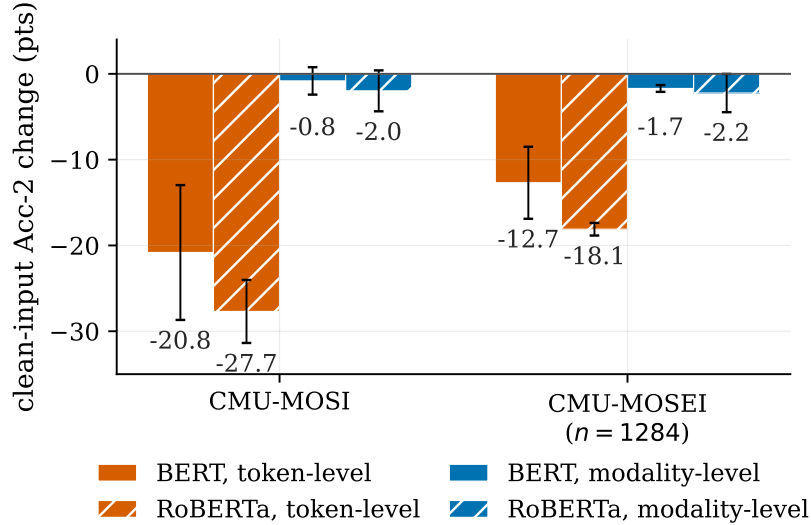
The encoder comparison uses model-specific tokenisation with consistent input casing. The benchmarks store BERT WordPiece indices, so we regenerate RoBERTa identifiers from the raw transcripts. Every CMU-MOSI transcript is upper-cased; we lowercase these inputs before applying RoBERTa’s case-sensitive tokenizer. Table 5 quantifies the result at the shared 50-token limit. On the training partition, lowercasing reduces mean retained RoBERTa length from 20.49 to 14.81 tokens; BERT averages 14.78. True truncation falls from 3.58% to 0.62%, compared with 0.55% for BERT. Here truncation means that the sequence exceeds the limit before clipping; sequences exactly at the limit are counted separately in the released audit. Lowercasing aligns length exposure more closely while retaining each encoder’s own tokenizer.

The clean-input protocol contrast persists with RoBERTa (Figure 3b). token-level erasure changes RoBERTa accuracy by -27.7 points on CMU-MOSI and -18.1 on size-matched CMU-MOSEI, while modality-level absence costs 2.0 and 2.2. RoBERTa begins from a higher clean baseline than BERT in this setting (0.853 against 0.835) and shows the larger clean-input change.

The size-matched results also retain a corpus difference: the change is 12.7 points on CMU-MOSEI and 20.8 on CMU-MOSI. Subsampling therefore accounts for part of the contrast between the full datasets while preserving variation associated with their content and feature distributions.



(a) Training-set size



(b) Text encoder and training protocol

Figure 3: **Data scale and text encoder modulate clean-input change.** (a) CMU-MOSEI subsampling; shading shows ± 1 standard deviation over three seeds at each subsample size and five at full size. Normalization uses the full training partition. (b) Mean paired-seed clean-input changes with ± 1 standard deviation: five seeds for BERT on CMU-MOSI, three otherwise. Both text encoders retain substantially more complete-input accuracy under modality-level absence.

Table 6: Clean-input performance under three interventions. Changing the training protocol to modality-level dropout (M1) preserves accuracy; freezing lower layers (M2) and reducing the encoder learning rate (M3) do not. Values are mean $\pm$ standard deviation over five baseline seeds and three intervention seeds; changes are paired by seed.

Dataset	Training scheme	clean-input Acc-2 $\uparrow$		change (pts)
		benign	text-scarce	
CMU-MOSI	token-level baseline	0.835 ± 0.007	0.627 ± 0.076	-20.8 ± 7.9
	M1: modality-level training	0.839 ± 0.006	0.837 ± 0.004	-0.2 ± 0.3
	M2: freeze embed. + 6 blk.	0.823 ± 0.007	0.586 ± 0.028	-23.7 ± 2.2
	M3: encoder LR 5×10^{-6}	0.833 ± 0.011	0.595 ± 0.026	-23.9 ± 3.7
CH-SIMS	token-level baseline	0.810 ± 0.011	0.764 ± 0.022	-4.6 ± 2.3
	M1: modality-level training	0.805 ± 0.009	0.814 ± 0.013	$+0.9 \pm 0.7$
	M2: freeze embed. + 6 blk.	0.828 ± 0.006	0.759 ± 0.029	-6.9 ± 2.4
	M3: encoder LR 5×10^{-6}	0.822 ± 0.002	0.770 ± 0.018	-5.2 ± 1.6

7. Mitigation

The modality-level training protocol keeps clean-input accuracy close to its benign-regime reference. Table 6 compares three interventions against the token-level baseline. Training with modality-level dropout (M1) moves the clean-input change from -20.8 points to -0.2 on CMU-MOSI and from -4.6 to $+0.9$ on CH-SIMS, while maintaining benign-regime accuracy. Freezing the embedding layer and the lower six transformer blocks (M2) and cutting the encoder learning rate fourfold (M3) do not provide the same recovery.

M1 changes the observation distribution and its associated reconstruction exposure. In this architecture, that complete training protocol retains clean-input accuracy more consistently than the two tested changes to encoder adaptation. Its clean-input result makes it a useful comparator when selecting a training protocol. Selection for deployment should additionally compare the candidates on a common suite of expected missing-input conditions.

8. Discussion and limitations

8.1. Implications for evaluation and training

The exposure measurement identifies two operational differences behind a shared nominal missing-rate schedule. Under text-scarce training on CMU-MOSI, modality-level dropout provides intact text in 30.33% of draws, compared with 0.04% for token-level erasure. It also supplies 1.070 rather than

0.033 reconstruction-eligible pairs per utterance (Table 1). Intact-input exposure and reconstruction exposure therefore vary together in the implemented comparison. These measurements explain what changes between protocols; the accuracy experiments measure the resulting system-level outcome. Identifying each exposure’s individual contribution requires independently controlling the reconstruction objective and the observation distribution.

The factorial control in Section Appendix A removes auxiliary reconstruction supervision while retaining the prediction architecture. At zero reconstruction weight, clean-input accuracy changes by -22.96 ± 8.56 points under token-level training and -1.10 ± 2.61 under modality-level training. Thus the observed protocol contrast also occurs without that auxiliary objective. Reconstruction-based imputation and the observation distributions still differ across protocols, so the control isolates loss-weight sensitivity rather than every consequence of missingness.

A practical evaluation should retain the clean test condition for every missingness-trained checkpoint. Reporting its absolute score, change from a specified training reference, and missing-input scores gives distinct views of retained capability and tolerance to unavailable inputs. This is especially useful when tuning corruption rates: a setting selected solely for performance on corrupted observations may give a different trade-off from one selected for a deployment mixture that includes complete inputs. The present results characterize the clean-input side of that trade-off; choosing a deployment policy also requires the expected test-time observation patterns.

The intervention results guide which training choices deserve direct testing. Reducing encoder plasticity through the two tested settings does not recover the token-level baseline, whereas changing the observation protocol does. This favors evaluating protocol choice early in model development. It also motivates a more targeted mechanistic experiment: vary corruption granularity while independently controlling intact-text exposure and imputation, then compare both complete-input outcomes and the representations along the text and fusion paths. Such a design would distinguish which components of the current operational difference contribute most to the measured change.

8.2. Scope of the evidence

The evidence is bounded to masked-language-model text pathways and one fusion architecture. The comparison evaluates complete training protocols, including their availability-gated reconstruction behaviour, rather than attributing the effect to a single layer or gradient path. Extending

clean-input performance to decoder-only text towers and published missing-modality architectures is the next step toward a broader protocol benchmark. The three-to-five-seed design supports descriptive comparisons; the rate-grid onset criterion and the single fixed subset at each sample size do not establish population-level thresholds or uncertainty over training-set selection.

9. Conclusion

We have compared missingness-training protocols through their clean-input changes relative to benign training. Across three sentiment benchmarks, high-rate token-level erasure changed complete-input accuracy by up to -27.7 points, while modality-level absence preserved substantially more of that capability in the shared reconstruction-based architecture. On the tested rate grids, token-level declines first pass two percentage points at 50–60% text erasure. Smaller training subsets generally show larger changes, and the effect appears with BERT, RoBERTa and [UNK] substitution. A separate five-seed control reproduces the protocol contrast at both tested reconstruction-loss weights, including zero. Together, these results establish clean-input performance as a consequential outcome of missingness-training protocol choice. Reporting the protocol alongside the nominal rate, and evaluating complete inputs alongside corrupted inputs, makes this distinction visible in multimodal evaluation.

Appendix A. Five-seed auxiliary-loss control

We test whether auxiliary reconstruction supervision is required for the clean-input protocol contrast. The study crosses two training protocols, two regimes (benign and text-scarce), two reconstruction-loss weights ($\lambda_{\text{rec}} = 0$ and 0.5), and five matched seeds, giving 40 runs on the full CMU-MOSI training partition. The architecture, batch size, learning rates, 40-epoch maximum and eight-epoch patience remain fixed. Checkpoint selection uses validation MAE under the training regime, averaged over two fixed mask draws. All five seeds enter every comparison. Reconstruction heads and the detached complete-input target forward pass remain active at both weights; the intervention removes the auxiliary objective, not imputation from the prediction pathway.

These runs use an RTX PRO 6000 Blackwell GPU, PyTorch 2.8.0 with CUDA 12.8, Transformers 4.57.6, BF16 autocast and scaled dot-product attention. Dataset and initial BERT-weight hashes match the local control inputs. The stable evaluation-seed implementation is shared across all 40 runs. This study is analysed separately from the archived experiments and the Metal pilot below; it does not isolate the effect of changing the historical validation-mask seed implementation from other runtime differences.

The clean-input contrast persists at both loss weights (Table A.1). At zero loss weight, token-level training changes Acc-2 by -22.96 ± 8.56 percentage points, compared with -1.10 ± 2.61 for modality-level training. At weight 0.5, the corresponding changes are -16.52 ± 5.09 and -0.61 ± 3.10 points. Token-level accuracy decreases in all five seeds at each weight. MAE gives the same ordering: increases of 0.553 versus 0.087 at weight zero and 0.458 versus 0.130 at weight 0.5 (Table A.2).

Table A.1: Complete-input Acc-2 in the separate CUDA factorial study. Values are mean $\pm$ population standard deviation over five seeds per condition. The last column uses within-seed text-scarce-minus-benign differences.

Training	λ_{rec}	Benign Acc-2 $\uparrow$	Text-scarce Acc-2 $\uparrow$	Δ_{clean} (pp)
token-level	0.0	0.833 ± 0.007	0.604 ± 0.080	-22.96 ± 8.56
token-level	0.5	0.829 ± 0.006	0.664 ± 0.051	-16.52 ± 5.09
modality-level	0.0	0.835 ± 0.005	0.824 ± 0.026	-1.10 ± 2.61
modality-level	0.5	0.836 ± 0.007	0.830 ± 0.026	-0.61 ± 3.10

For each seed, we subtract the modality-level clean-input change from the token-level change. This protocol contrast is -21.86 ± 11.16 points at weight zero and -15.91 ± 6.25 at weight 0.5; it is negative in every seed at both

Table A.2: Complete-input MAE for the same 40 CUDA runs, reported as mean $\pm$ population standard deviation over five seeds per condition. Changes are computed within seed; positive changes indicate greater prediction error.

Training	λ_{rec}	Benign MAE $\downarrow$	Text-scarce MAE $\downarrow$	MAE change $\downarrow$
token-level	0.0	0.793 ± 0.036	1.346 ± 0.105	$+0.553 \pm 0.117$
token-level	0.5	0.810 ± 0.012	1.268 ± 0.071	$+0.458 \pm 0.068$
modality-level	0.0	0.752 ± 0.011	0.840 ± 0.054	$+0.087 \pm 0.053$
modality-level	0.5	0.736 ± 0.008	0.866 ± 0.087	$+0.130 \pm 0.094$

weights. The coefficient interaction, defined as the latter contrast minus the former, is $+5.95 \pm 14.65$ points. Its seed-level values range from -19.36 to $+22.41$ points, so the mean attenuation does not imply a consistent benefit from the auxiliary objective. The MAE contrast is 0.466 ± 0.169 at zero weight and 0.327 ± 0.085 at weight 0.5, with an interaction of -0.138 ± 0.215 . The stable result is that the protocol contrast remains when auxiliary reconstruction supervision is removed. The remaining protocol differences include the observation distribution and availability-dependent prediction path within this architecture.

Appendix B. Supplementary Metal pilot

We examine auxiliary-loss sensitivity with a matched single-seed control on the full CMU-MOSI training partition. Both runs use token-level text-scarce training, seed zero and the same reconstruction architecture, with $\lambda_{\text{rec}} = 0$ or 0.5. The reconstruction heads remain in the prediction path, and both runs retain the complete-input target forward pass. The intervention changes the auxiliary loss coefficient while preserving those computations. Validation uses deterministic masks under the training regime, with the learning rates, batch size and stopping rule of Section 4.

The pilot uses PyTorch 2.14.0 and Transformers 4.57.6 on Apple Metal, with full-precision computation and eager BERT attention to support attention dropout. We report this local study separately from the archived CUDA results. Both runs stop after 12 epochs and select the fourth epoch by validation MAE. Table B.1 reports their complete-input outcomes. Removing the auxiliary loss gives 54.42% accuracy, compared with 57.32% at weight 0.5; MAE is 1.415 and 1.381, respectively. Thus loss removal does not improve clean-input performance for this seed and training condition. This pilot measures auxiliary-loss sensitivity at a fixed text-scarce regime.

The complete five-seed, two-regime comparison is reported separately in Section Appendix A using a consistent CUDA runtime.

Table B.1: Single-seed CMU-MOSI auxiliary-loss pilot under token-level text-scarce training. Acc-2 and MAE use complete test inputs. Epoch indices are one-based. Both conditions use the same local Metal runtime and reconstruction architecture.

Rec. coefficient	Clean Acc-2 $\uparrow$	Clean MAE $\downarrow$	Selected epoch
0.0	0.544	1.415	4
0.5	0.573	1.381	4

We additionally evaluate both selected checkpoints on common missing-input suites (Table B.2). Each suite uses the text-scarce rate distribution, five deterministic mask repetitions and the same test ordering and batch size for both checkpoints. Under token-erased test inputs, mean accuracy is 55.21% at weight zero and 58.26% at weight 0.5. Under whole-modality dropout, both obtain 53.90% accuracy, with MAE 1.435 and 1.423, respectively. The coefficient comparison therefore depends on the evaluation condition. Both models in this pilot were trained with token-level erasure; the suites compare auxiliary-loss settings under shared test conditions rather than comparing the two training protocols.

Table B.2: Common-suite evaluation of the two pilot checkpoints. Both models were trained with token-level text-scarce missingness. Each test operator uses the text-scarce nominal rate distribution; values are means over five mask repetitions, not repeated training seeds.

Rec. coefficient	Test operator	Acc-2 $\uparrow$	MAE $\downarrow$
0.0	token-level	0.552	1.412
0.0	modality-level	0.539	1.435
0.5	token-level	0.583	1.371
0.5	modality-level	0.539	1.423

CRedit authorship contribution statement

Zehao Li: Methodology, Software, Formal analysis, Writing – original draft, Writing – review & editing.

Declaration of competing interest

The author declares no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

All datasets used in this study are publicly available. CMU-MOSI [19], CMU-MOSEI [20] and CH-SIMS [21] were obtained as the pre-extracted unaligned feature files distributed by the MMSA toolkit; the SHA-256 digest of every file used is recorded in the released code and checked before use. The manuscript source, experiment code, per-run results and reproduction instructions are available in the public repository.

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